Software Requirements Specification

for

Text to Insight: Summarise & Respond (TSR)

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Revision History

Date	Change	Reason for Changes	Mentor Signature

Introduction

1.1 Purpose of the Project

The task of extracting valuable insights from vast textual datasets has been a longstanding challenge. Various other approaches to text summarization such as rule based techniques or extractive techniques have been used, while the question-answering systems relied on structured data sources or manually curated knowledge bases. Therefore, the problem we aim to address in this project is to develop a solution that capitalizes on the capabilities of LLMs, thereby redefining the boundaries of what is achievable in document understanding by text summarization and querying over the given textual information.

The purpose of the project is to create a Text Summarization System for concise and informative summaries from lengthy text content while maintaining high accuracy. Build an Insightful Response Generation system capable of generating insightful responses to user queries. Evaluate the effectiveness and usability of the summarization and insight generation processes.

1.2 Target beneficiaries

The prime beneficiaries of this project encompass a diverse range of stakeholders. Researchers and academics stand to gain a powerful tool for expediting the extraction of critical information from extensive textual resources. This will notably enhance their research efficiency and deepen their comprehension. Likewise, content creators and curators will find immense value in the system's ability to swiftly distill large volumes of text into concise, informative content suitable for various platforms. Information seekers, whether professionals or individuals, will benefit from the system's capacity to provide insightful responses to their queries, significantly streamlining their search process. Businesses and enterprises can leverage this tool for expeditious document analysis, facilitating the rapid extraction of relevant insights crucial for informed decision-making. Educational institutions, too, can employ this system as a valuable learning aid, aiding students in understanding and summarizing complex textual materials. Lastly, general users grappling with substantial volumes of text, such as articles, reports, or documents, will find this system invaluable for swiftly obtaining summarized content and pertinent responses to their queries

1.3 Project scope

The project aims to develop a Text Summarization & Insightful Response (TSR) system utilizing advanced Large Language Models (LLMs). This system targets the enhancement of document understanding by generating concise, informative summaries from extensive text and providing insightful responses to user queries. The objectives include creating a robust summarization tool and a responsive QA system, ultimately improving information retrieval. The deliverables will consist of the fully functional TSR software, along with documentation and user guides. The project's scope extends to applications in fields such as content curation, research, and information extraction for improved efficiency and comprehension.

2. PROJECT DESCRIPTION

2.1 Reference Algorithm

Our Text Summarization and Insightful Response project relies on the MapReduce framework and retrieval augmented generation. The process begins with data acquisition from a variety of sources, such as text documents, web pages.Text summarization is a process of grouping key points of the text from a large collection of text sources[1]. The collection of sources can be reviewed from the web, documents, blog posts, social media posts, news feeds, etc. Generating a concise and precise summary of voluminous texts is the key idea in summarization[2]. Subsequently, data preprocessing is performed to remove the irrelevant information, like punctuation and special characters, and to structure the text for compatibility with MapReduce. The Map phase follows, where data is transformed into key-value pairs, these keys represent specific attributes of the text, and the values contain corresponding text segments. Mappers handle tasks such as word counting, sentence segmentation, and feature extraction in preparation for text summarization. Following the Map phase, the Shuffle and Sort phase is initiated to arrange the key-value pairs, grouping related data and organizing it for the Reduce phase. The Reduce phase is where the main work happens. Summarization methods may encompass extractive or abstractive approaches, involving sentence ranking and content selection. Insightful response generation leverages natural language processing techniques and context-awareness to create responses that are informative and contextually relevant from the ingestion of documents into a queryable format and then the retrieval augmented generation chain.

2.2 Data

The Scientific Paper Dataset contains two sets of long and structured documents. The datasets are obtained from ArXiv and PubMed OpenAccess repositories.
Both "arxiv" and "pubmed" have two features:

- article: the body of the document, paragraphs separated by "/n".
- abstract: the abstract of the document, paragraphs separated by "/n".
- section_names: titles of sections, separated by "/n".

2.3 SWOT Analysis

Strengths:
- Efficient data processing using MapReduce.
- Produces high-quality summaries and responses.
- Versatile and scalable for various text sources.

Weaknesses:
- Complex setup and resource-intensive operation.
- Limited suitability for real-time applications.

Opportunities:
- Increasing demand in the market.
- Potential for integration with AI and analytics tools.

- Customization options for users.

Threats:

- Competition and evolving technology landscape.
- Potential impact from data privacy regulations.
- Maintaining user trust in output quality.

2.4 Project Features

- **Text Summarization Engine:** Developing a text summarization so that it can condense a lengthy part of text into a shorter with relevant information and its main points.

- **Insightful Response:** Implement a response generation system which is capable of providing valuable insights and relevant responses to user queries.

- **Metrics and Evaluation:** Incorporate metrics and evaluation tools to assess the accuracy, conciseness, and informativeness of generated summaries and responses.

2.5 User Classes and Characteristics

- **End Users:** Non-technical individuals, prioritize user-friendly experience and easily comprehensible summaries and responses.

- **Data Analysts/Researchers:** Technical experts, seek algorithm control and customization for specific research requirements.

- **System Administrators:** IT professionals responsible for system setup, maintenance, and security monitoring.

- **Feedback Contributors:** Users provide valuable feedback to enhance the system's output quality and performance.

2.6 Design and Implementation Constraints

Installation and configuration of any open-source IDE for the PYTHON programming language are included in the setup and maintenance. Other than those defined for an application that executes according to the standard documentation, the IDEs have no extra requirements. Unless an IDE is cloud-based, a developer's operating system will limit which IDEs are accessible, and if the application is meant for an end-user with a certain operating system this may be an additional constraint.

2.7 Design Diagrams

- **Flowchart:**
  Following is the flow chart in fig 1:

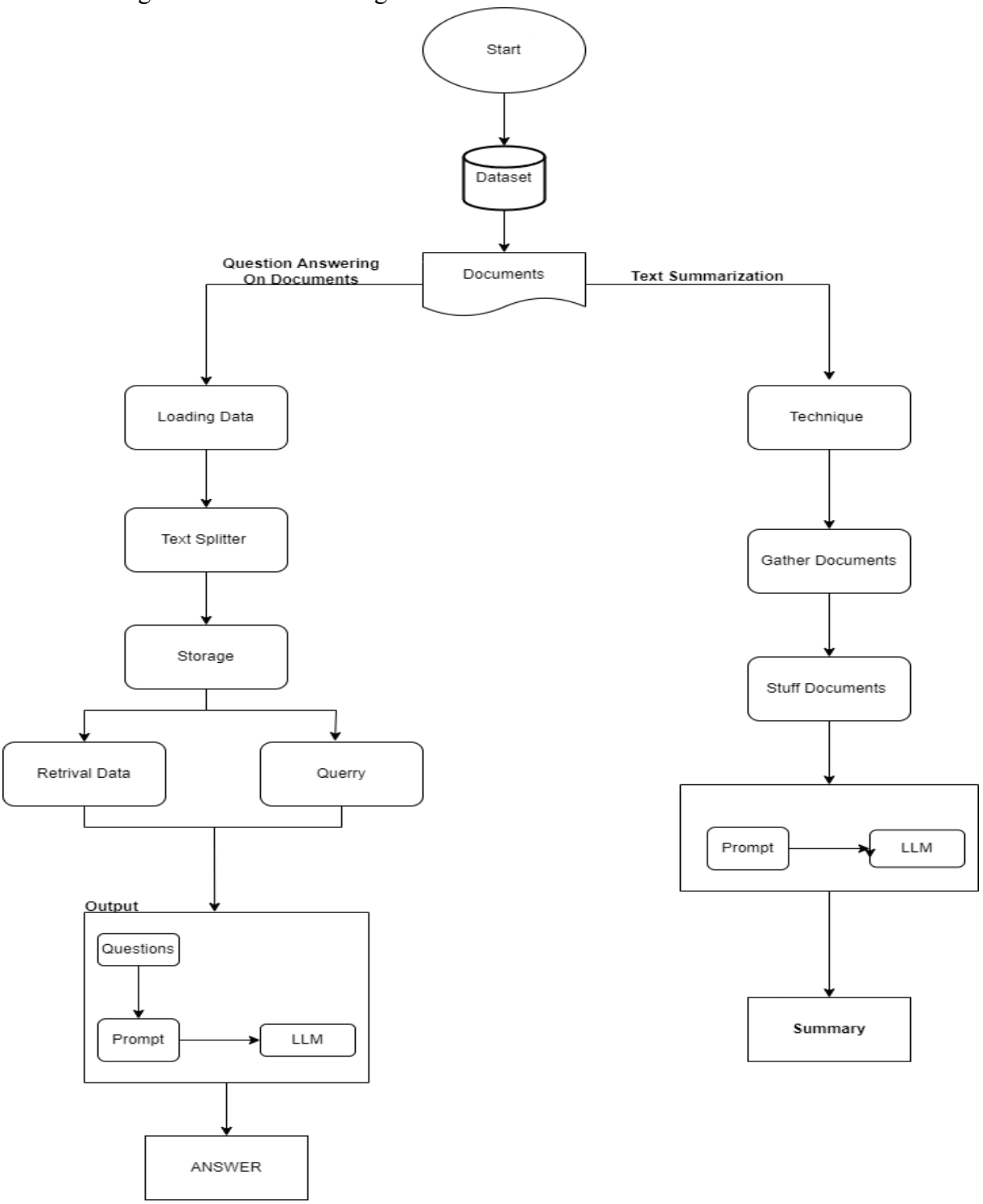


Figure 1: Flowchart for the system

- **Use Case Diagram:**
  According to Fig. 2, it expresses what are the main functions our system provides. Use Case Diagram for our system:

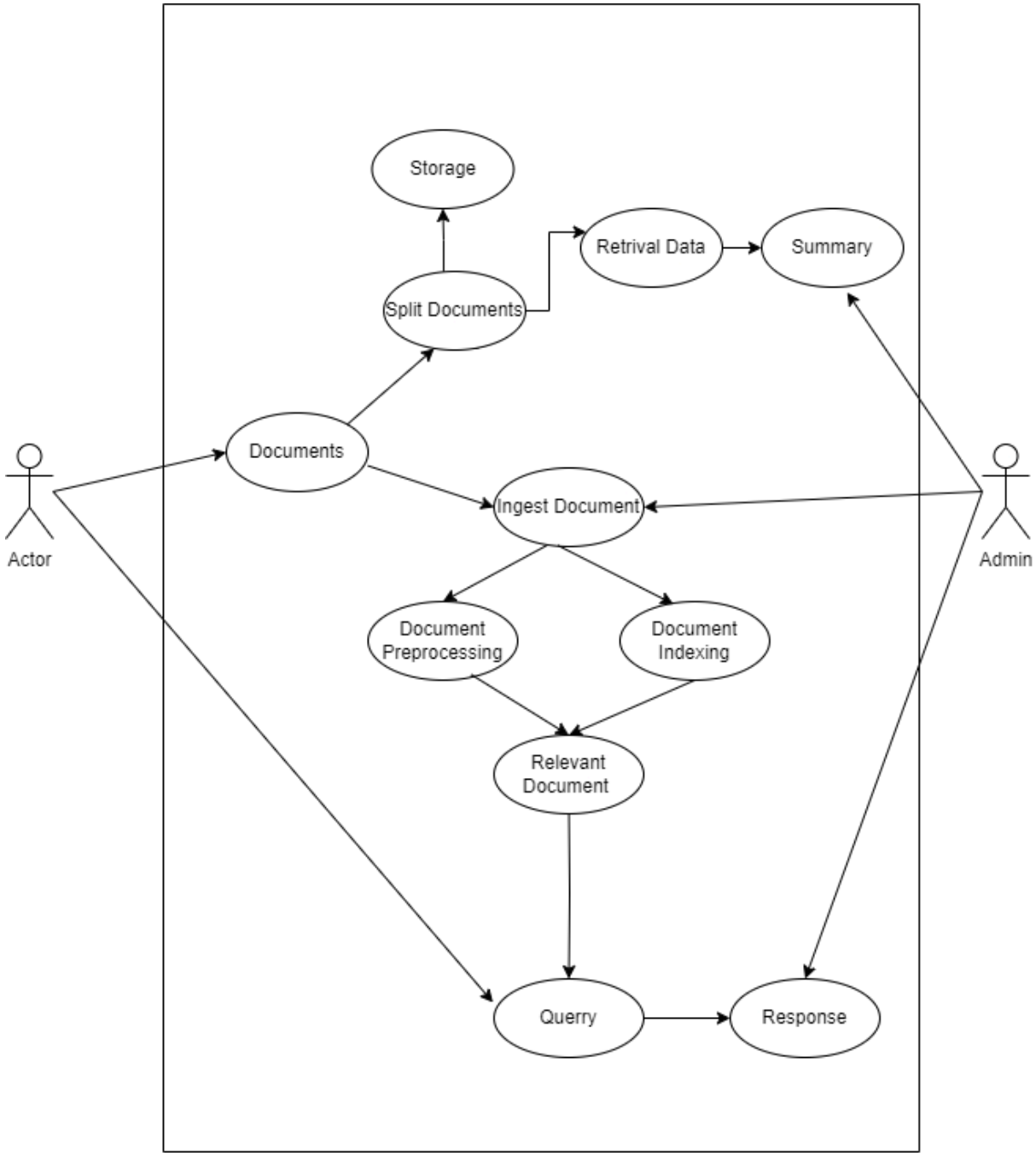


Figure 2: Use Case Diagram of the system

2.8 Assumption and Dependencies:

At this time, no particular assumptions or dependencies are being considered

3. System Requirements

3.1 User Interface

The user can interact with our project via our GUI.

3.2. Software Interface

The software used to implement the codes

- Operating System - Windows/Linux
- Python IDE(PyCharm, Visual Studio,etc)

3.3. Database Interface

Not Applicable

3.4. Protocols

All standard protocols could be used as an integrated part of the OS but direct .

4. NON- FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements:

- **Summarization Accuracy:** It defines specific metrics to measure the accuracy of our system. It includes metrics like BLEU scores and ROUGE scores.
- **Conciseness:** Our model ensures that the generated summaries are concise and does not contain any unnecessary information.
- **Informativeness:** Summaries must convey the important content from the original text. Defining the metrics for measuring the content from the summaries .
- **Response Insightfulness:** Develop metrics to evaluate the valuable insights provided by our model.

4.2 Security requirements:

We are not directly responsible for the security of the user's data as we use a cloud based service for dealing with sensitive data.

4.3 Software Quality Attributes

- **Accuracy:** Our model will provide accurate results for both text summarization and response generation systems. As high inaccuracy in summaries or responses may lead to inaccurate information.
- **Reliability:** Our software must be dependable and produce accurate results. Users should rely on the system's output for their information needs.
- **Scalability:** Handling a wide range of data sizes and user demands.

- **Performance:** Optimizing the performance of our system to generate summaries and responses quickly and efficiently.
- **Robustness:** Making the system robust for unexpected inputs or errors. It must handle all the critical cases and produce effective results.
- **Maintainability:** Ensure that your codebase is well-organized, documented, and easy to maintain. This will make it easier to adapt and extend your system as needed.
- **Security:** Addressing security concerns, We are not directly responsible for the security of the user's data as we use a cloud based service for dealing with sensitive data.
- **Usability:** A user-friendly interface that makes it simple for users to interact with the system and provide feedback can be designed into the system. We could also think about incorporating the system into existing music streaming platforms, which would make it easier to use and more popular.
- **Reusability**: The system can be designed as a modular system with distinct components for user feedback, music recommendation, and facial recognition. The system can be easily adapted to various use cases or integrated into other applications in this manner.
- **Adaptability**: Over time, the system should be flexible and easily adaptable to different domains or datasets. Through user interactions, the system could constantly learn and adjust its suggestion algorithms.
- **Interoperability:** To offer a seamless user experience, the system should be able to integrate with the model for summaries and responses. Users might be able to save the summaries or the responses for instance.
- **Portability** refers to how easily it can be used with various hardware and running systems.

5. OTHER REQUIREMENTS

There are no other requirements for our proposed model.

6. References

[1] Summarizing large text collection using topic modeling and clustering based on MapReduce framework; *N K Nagwani;*Nagwani Journal of Big Data (2015) 2:6 DOI 10.1186/s40537-015-0020-5.

[2] Computational Techniques for Text Summarization based on Cognitive Intelligence; *V. Priya and K. Umamaheswari (2023)*

APPENDIX A: GLOSSARY

- **LLMs (Large Language Models):** Refers to Large Language Models, which are AI models designed to understand and generate human-like text. Examples include GPT-3 and similar models.
- **Text Summarization:** The process of condensing a large part of text into a shorter, with relevant information and its main points.
- **Insightful Response Generation:** The generation of responses to user queries that provide valuable and meaningful insights based on the content of the input text.

- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** A set of metrics commonly used to evaluate the quality of text summarization by comparing the overlap of n-grams (words or phrases) between the generated summary and reference summaries.
- **BLEU (Bilingual Evaluation Understudy):** A metric used to assess the quality of machine-generated text, including text summarization, by comparing n-grams in the generated text with those in human-generated reference texts.
- **Graphical User Interface (GUI):** The graphical or command-line interface through which users interact with the system.
- **API (Application Programming Interface):** A set of rules and protocols that allows different software applications to communicate with each other.
- **OS:** An operating system is the most important software that runs on a computer. It manages the computer's memory and processes, as well as all of its software and hardware.
- **IDE:** An integrated development environment (IDE) is a software application that provides full software development features to computer programmers. A source code editor, build automation tools, and a debugger are usually included in an IDE.
- **Python:**Python is a high-level, interpreted programming language that is used for a wide range of applications.

APPENDIX B: ISSUES LIST

- **System Failure:** A system failure can occur because of a hardware failure or a severe software issue, causing the system to freeze, reboot, or stop functioning altogether.
- **Runtime-error:** Run-time errors are errors that occur during program execution (run-time) following successful compilation. These errors are difficult to locate since the compiler does not indicate the line where the error occurs.
- **Compile-time error:** Compile-Time errors are errors that arise when you break the conventions of writing syntax. Before the code can be compiled, this compiler error indicates that something needs to be fixed. All of these problems are known as compile time errors since they are identified by the compiler.